Fraction of volume Submerged = $\frac{\text{Density of obj}}{\text{Density of Liquid}}$

Pg① $\gamma_{Glycerin} > \gamma_{kerosene} > \gamma_{water}$

Fluid & DYNAMICS

Fluids are substances that can flow i.e liquid, gases (plasma)

gases flow → Aerodynamic

Liquid flow → Hydrodynamic

Ideal Fluid :

- Irrotational flow (no motion other than translational).
- Steady (laminar / Steam line) (uniform flow) velocity = same.
- non viscous, no internal friction force, $V_{rel} = 0$
- Incompressible (ρ = constant) liquids not gases

TYPES OF Fluid Flow

Steady flow (laminar)

- Smooth & orderly
- All points of a layer passes through same path & velocity.
- Stemline flow don't cross each other.
- velocity along layer remains constant.
- Speed at low Speed.

↓

gentle breeze

Turbulent flow

- Irregular & non-Steady.
- continuously pressure & velocity fluctuation.
- fluid's velocity exceeds a certain critical velocity. ($> 0.16 m/s$)
- forming whirlpools

↓

- flow of waters from top of mountain.
- water flow at Sea Shore.

Pg ②

Reynold's number,

"Nature of flow."

It is the ratio between constant inertial force per unit area to viscous force per unit area.

$$N_R = \frac{\text{inertial force}/A}{\text{viscous force}/A}$$

0 —— 2000 laminar] → between it there occurs

> 3000 Turbulent] transition when flow convert from laminar to turbulent.

Viscosity, → not a physical quantity

coefficient of viscosity is a physical quantity.

↓

fluid vs fluid opposition

Newton's law of viscous force, Jab aik layer dosri layer ke upar slide karti hai to ain dono ke darmiyan tangential viscous force present hoti ha

F_t

η = coefficient of viscosity

$$\eta = \frac{F_t}{A \frac{\Delta V}{\Delta X}}$$

"viscous force acting per unit area with unit velocity gradient"

$$\eta = \frac{N \cdot s}{m^2} = Pa\,s$$

SI unit of viscosity

↓

poiseuille

$Pa\,s = Nm^{-2}s$

$F_t \propto A$ (plane area)

$F_t \propto \Delta V / \Delta X$ (gradient)

↓

relative velocity w.r.to distance between layers.

$$F_t = (-)\eta A \Delta V / \Delta X$$

↓

shows force is opposite to velocity viscous force.

~~Jab bhi kisi velocity ki~~

Jab bhi kisi velocity ki change dekhi jaye w.r.to distance, use gradient kehte hain.

In CGS System:-

η unit = dyne cm^{-2} s

poise.

1 Pa s = 10 poise

1 poiseuille = 10 poise

For Spec

Pg 3

~~effect~~ $\eta_{liquid} = 100\ \eta_{gases}$

FACTORS OF DEPENDENCE:

(a) NATURE OF FLUID:

Liquid $\rightarrow \eta \propto$ intermolecular forces
Gases $\rightarrow \eta \propto$ rate of diffusion.

(b) Temperature:

Liquids $\downarrow \eta \propto 1/temp \uparrow$.

Gases $\uparrow \eta \propto$ ~~1/~~ temp$\uparrow$

DRAG FORCE: "Solid vs fluid opposition"

at rest there won't be any drag force.

Drag force is supported by Stokes law.

$$F_d = 6\pi \eta r v$$

Applicable to only spherical objects. as $\boxed{F_d \propto r}$

$F_d \propto \eta$

$F_d \propto r$ (radius of object)

$F_d \propto v$ (velocity of object)

Limitation:- only applicable to (Stokes law)

Spherical objects

Slow moving bodies.

Note:- at high speed: $F_d = \frac{1}{2} \rho C_d A v^2$

C_d → drag coefficient

TERMINAL VELOCITY

in vacuum $V_T = 0$

Applicable when an object moves in a fluid & its $a=0$ that constant velocity is called terminal velocity.

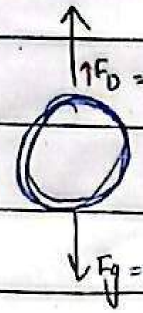
"Maximum uniform velocity at which acceleration of the object becomes zero".

Pg (4)

here we usually ... up thrust force. Air density is negligible but we can't ignore ... it ... in ...

when you displace liquid or gases a force is applied on you that is called up thrust.

we neglect F_{thrust}



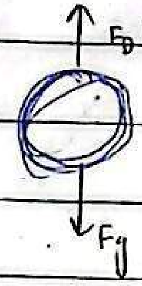
$\boxed{\text{before}}$ $a < g$

$F_{net} = F_g - F_D$

$ma = mg - 6\pi\eta r v$

$a = g - \frac{6\pi\eta r v}{m}$

that's why $a < g$



$F_{net} = 0$ $\boxed{\text{At}}$

$a = 0$

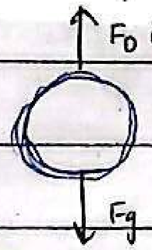
$v \longrightarrow v_T$

$F_{net} = F_g - F_D$

$F_{net} = 0$

$0 = ma$

$a = 0$

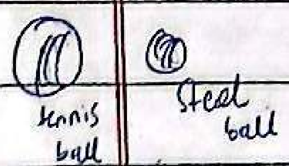


Mathematical Representation:

$$\boxed{V_t = \frac{mg}{6\pi\eta r}}$$

Applicable when m = constant.

$V_t \propto 1/r$ ①



$r_t > r_s$

$V_t < V_s$

$\Rightarrow \rho = M/V \Rightarrow m = \rho V = \rho\left(\frac{4}{3}\pi r^3\right)$

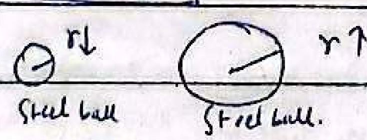
$$\boxed{V_t = \frac{2\rho g r^2}{9\eta}}$$

$\rho \rightarrow$ density of object.

Limitation: Applicable when ρ = constant.

we have ignored density of fluid as we know air density is negligible.

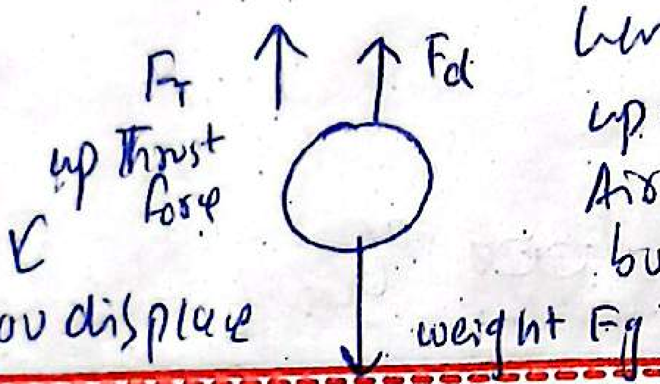
two rain droplets we will use this formula.



$V_t \propto r^2$

ρ = equal.

$m \neq$ equal.



here we usually ... upthrust force because Air density is negligible but we can't ignore ... thrust force in water

when you displace liquid or gases a force is applied on you that is called upthrust

we neglect F_{thrust}

$\pi \eta r v$ ↑ velocity increases

$F_{net} = F_g - F_D$

$ma = mg - 6\pi\eta r v$

$\boxed{\text{before}}$ $a < g$

$a = g - \frac{6\pi\eta r v}{m}$

that's why $a < g$

stant

crease) → max. = F_g

$F_{net} = 0$ $\boxed{\text{At}}$

$a = 0$

→ v_T

$F_{net} = F_g - F_D$

$F_{net} = 0$

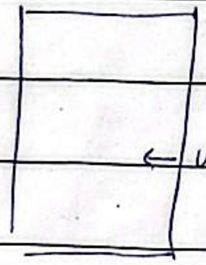
$0 = ma$

(5) if we have a fluid whose density can't be ignored.

$$\boxed{V_t = \frac{2(\rho - \sigma) g r^2}{9\eta}}$$ density of fluid.

_______ :Ext

(4) Steel ball



we will use this formula. because we will also ~~~~ include thrust force concept.

Note: Lighter bodies tend to gain their terminal velocity earlier than heavier bodies e.g. rain droplet.

Lighter body: Smaller weight mg. Needs less drag force to balance weight. That drag is reached at lower speed. So acceleration becomes zero sooner.

Dependence of terminal velocity:

(i) radius → $V_t \propto 1/r$ ∵ m = constant

→ $V_t \propto r^2$ ∵ ρ = constant

(ii) density.

density of object — $V_t \propto \rho$

density of fluid — $V_t \propto (\rho - \sigma)$

if $\rho_o > \sigma_f$ ⟶ $+V_t$ object moves downward.

if $\rho_o < \sigma_f$ ⟶ $-V_t$ object moves upward.

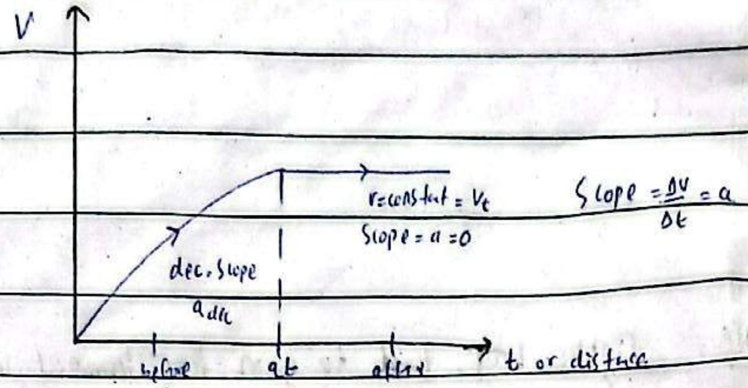
- fizzy drinks bubbles moves upward.
- clouds in sky.

if $\rho_o = \sigma_f$ ⟶ $V_t = 0$

water in water.

(iii) viscosity of fluid. $V_t \propto 1/\eta$

Graphical Representation:



Terminal velocity of a Paratrooper:

Paratrooper achieve its terminal velocity twice:

Apna parashoot open krne se pahale | Apna parashoot open karne ke bad.

both terminal velocities are different

1- accelerates as he/she begin to fall. as v↑

2- Speed up (v inc) the F_D will increase.

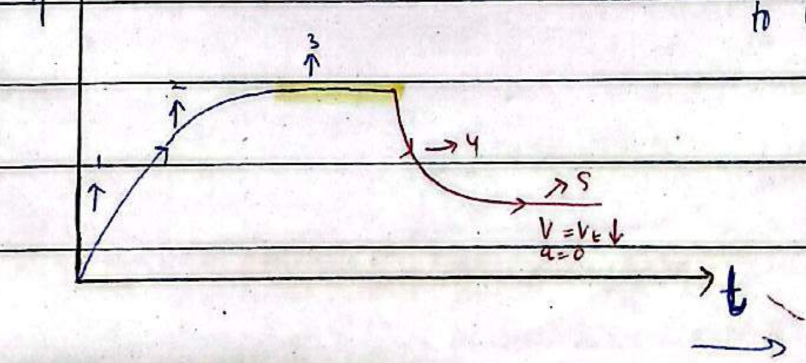
3- At terminal velocity, a = 0, $F_D = w$; $F_{net} = 0$

Parashoot has not been opened till now.

4- Now parashoot opens which increase the drag force & slow down the paratropper.

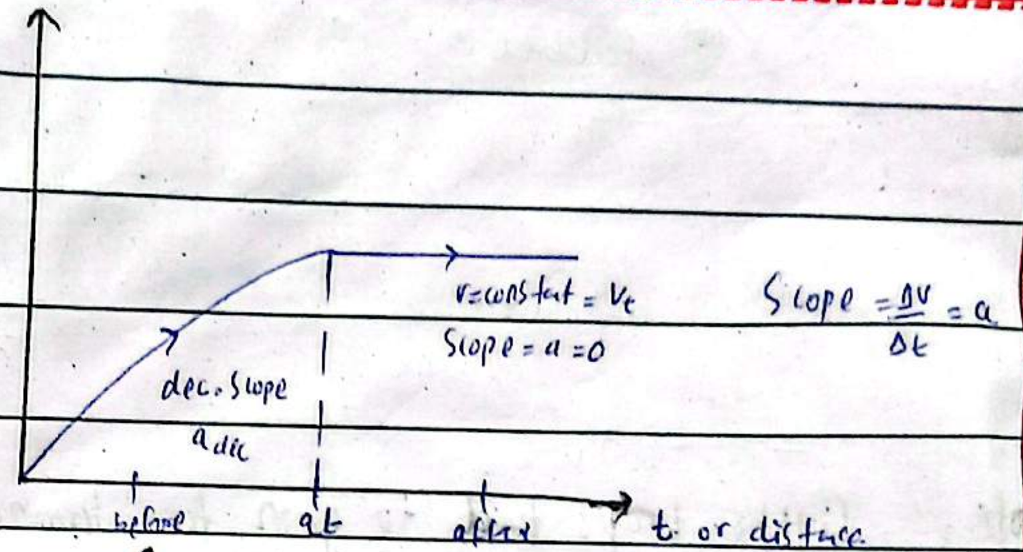
5- Paratrooper continues to slow down until the new terminal velocity is achieved.

As soon a parashoot opens, surface area has increased, Drag force value will increase paratropper will decrease its speed to make it (Fd) equal to weight



Graphical Representation:

:تاریخ



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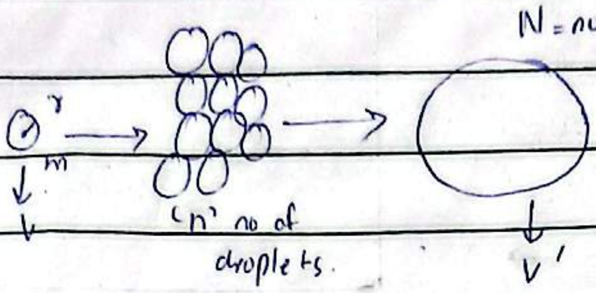
2- Speed up (v inc) the F_D will increase.

Pg 7 Note:

"N" no of droplets $V_N = N^{2/3} V$

N = no. of droplets.



mass has increased n times so V_t also changes.

EQUATION OF CONTINUITY:- ideal fluid flow.

"Acc. to law of conservation of mass"

for liquid: "The Product of cross-sectional area of the Pipe & the fluid Speed at any point along the Pipe is a constant equal to volume flow rate" i.e $AV = V/t$] Liquids density is constant

gases as their density is not constant: "Mass flow rate at any point during fluid flow is constant & equal to product of density, cross sectional area & speed of fluid. i.e $\rho AV = m/t$

" Speed of fluid through a Pipe is inversely proportional to cross sectional area i.e $V \propto 1/A$.

MATHEMATICAL REPRESENTATION.

For compressible fluids (non-ideal)

$\rho AV = \text{const} = m/t \ kg/s$

$\rho_1 A_1 V_1 = \rho_2 A_2 V_2$

For incompressible fluids. (ideal)

$AV = \text{constant} = V/t \ m^3/s$

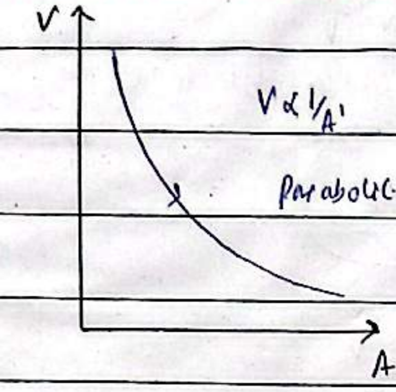
$A_1 V_1 = A_2 V_2$

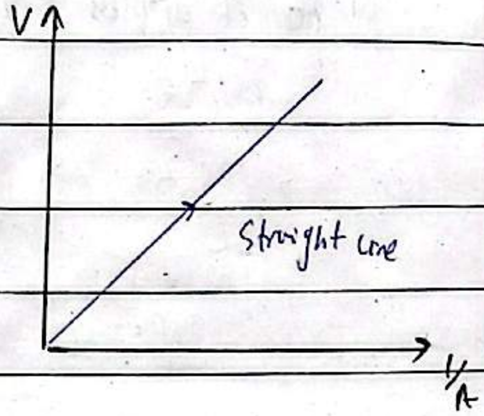
$r_1^2 V_1 = r_2^2 V_2$

$d_1^2 V_1 = d_2^2 V_2$

Graphical Representation:

Pg (8)





Bernoulli's Equation:

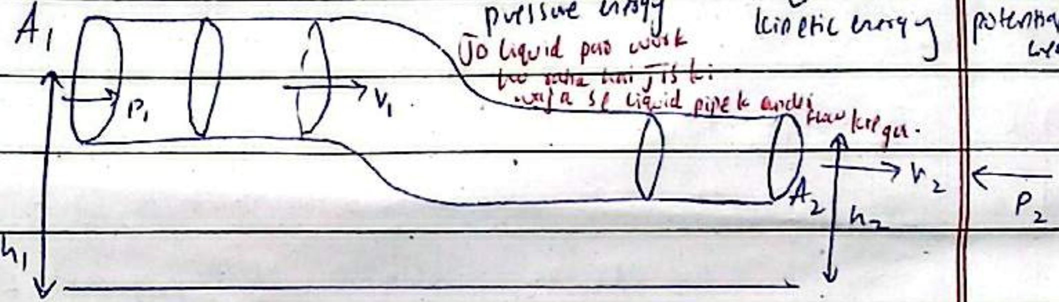
"Law of conservation of energy"

There are three types of energies

basically work ← pressure energy — Jo liquid par work ho raha hai jis ki wajah se liquid pipe k andar flow krega.

kinetic energy

potential energy



"Bernoulli equation is the relation between speed, pressure & height for an ideal fluid flow"

"The total energy (Pressure, potential & kinetic energy) per unit volume of an incompressible, steady & non-viscous fluid remains constant throughout the flow".

MATHEMATICAL REPRESENTATION:

pressure energy ← PV ; K.E → $\frac{1}{2}mv^2$; P.E → mgh

$$PV + \frac{1}{2}mv^2 + mgh = const. \quad \text{—(i)}$$

dividing by volume V

General eq/ 2

$$P + \frac{1}{2}\rho v^2 + \rho g h = const. \longrightarrow$$

Bernoulli equation is formed when we divide energy eq with volume.

Bernoullis equation per unit volume.

dividing by mass eq (i)

$$\frac{P}{\rho} + \frac{1}{2}v^2 + gh = const. \quad \text{(ii)} \longrightarrow$$

Bernoullis eq per unit mass.

dividing by "g"

$$\frac{P}{\rho g} + \frac{v^2}{2g} + h = constant.$$

pressure head ← $\frac{P}{\rho g}$; velocity head ← $\frac{v^2}{2g}$; h → gravitational head.

Learning constant's units.

تاریخ:

Pg 9 $PV + \frac{1}{2}mv^2 + mgh = const. \rightarrow$ Joule

all are energies.

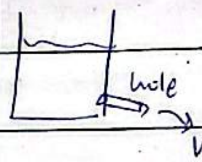
$P + \frac{1}{2}\rho v^2 + \rho g h = const.$ Pascal.

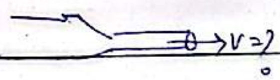
$\frac{P}{\rho} + \frac{1}{2}v^2 + gh = const. \rightarrow m^2/s^2$

$\frac{P}{\rho g} + \frac{v^2}{2g} + h = const \rightarrow m$

Applications of Bernoulli's Equation:

- Torricelli's Theorem (determination of Speed of efflux)
- Relation btw Static Speed & pressure — $P \propto 1/v$
  - ex: Magnus effect, chimney, carburettor, Bunsen burner, Lift of Aeroplane
- Venturi ~~Relation~~ (Speed of fluid)
- Blood flow.

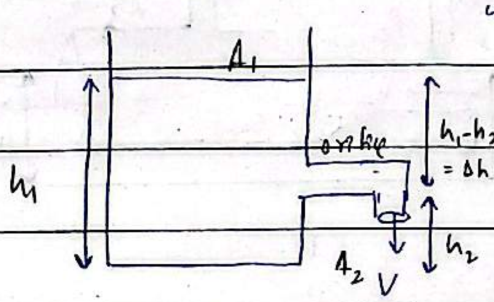




Torricelli's Theorem:

"tellas us how fast a fluid is flowing through an orifice or Spigot (Small hole)".

"The Speed of efflux is equal to velocity gained by the fluid while falling through a distance equal to difference of heights under the action of gravity"



MATHEMATICAL form

$$V = \sqrt{2g\Delta h}$$

$$V = \sqrt{2g(h_1 - h_2)}$$

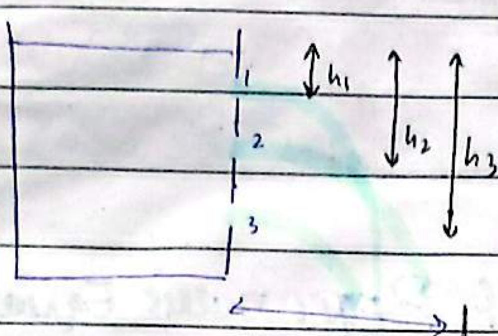
Pg 10 Conditions :

$A_1 >>> A_2$ ________ :है

$V_1 <<< V_2$

$V_1 \simeq 0$

$P_1 = P_2$ (atmospheric pressure)



Range $= 2\sqrt{h \times H}$

height of orifice from ground

distance from orifice to water level.

Speed of the water will be maximum for hole # ________

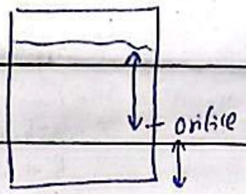
Range of the water will be maximum for hole # 2 (midpoint)

$V = \sqrt{2gH}$ → distance from orifice to water level.

$H_3 > H_2 > H_1$

$V_3 > V_2 > V_1$

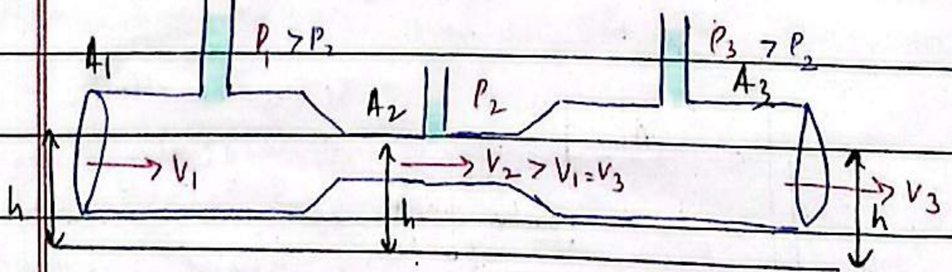
$R = 2\sqrt{h \times H}$



trick at midpoint product of both h & H would be greater.

Relation between Speed & pressure:

Speed of fluid is inversely proportional to the Pressure.



$$A \alpha P \propto 1/V$$

$$P_1 + \tfrac{1}{2}\rho V_1^2 + \cancel{\rho g h_1} = P_2 + \tfrac{1}{2}\rho V_2^2 + \cancel{\rho g h_2}$$

$h_1 = h_2$

$$\boxed{P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2} = \text{const}$$

→ Jaha par 2 chezon ka sum constant hota hai waha par unka relation inverse hota hai.

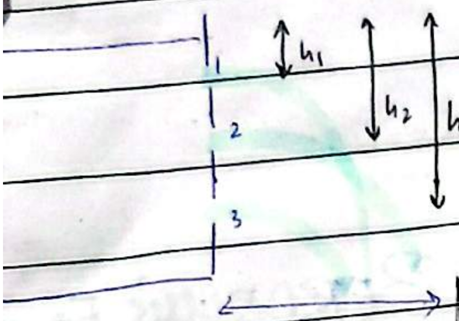
Conditions :

$A_1 >>> A_2$

$V_1 <<< V_2$

$V_1 \simeq 0$

$P_1 = P_2$ (atmospheric pressure)



Range $= 2\sqrt{h \times H}$

h → height of orifice from ground

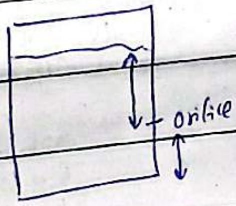
H → distance from orifice to water level.

...peed of the water will be ...m for hole # ________

Range of the water will be maximum for hole # 2 (mid point)

$R = 2\sqrt{h \times H}$ → trick at midpoint product of both h & H would be greater.

$V = \sqrt{2gH}$ → distance from orifice to water level.

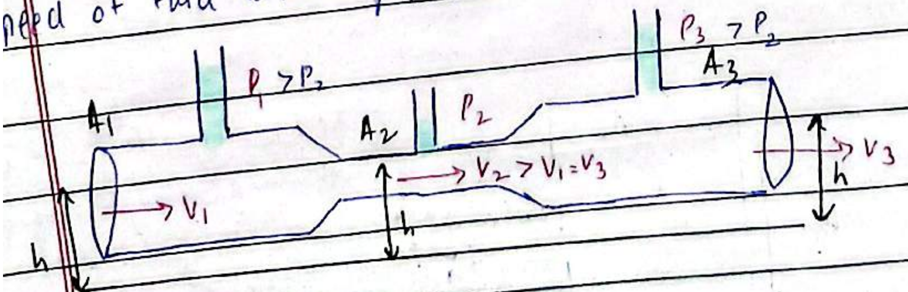


$H_3 > H_2 > H_1$

$V_3 > V_2 > V_1$

...lation between Speed & pressure:

...peed of fluid is inversely proportional to the Pressure.



$A \propto P \propto 1/v$

$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2$

$h_1 = h_2$

$\boxed{P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2}$ = const

→ Jaha pars 2 chezon ka sum constant huta hai waha par unka relation inverse huta hai.

$P + \frac{1}{2}\rho v^2 = const \longrightarrow P \propto \frac{1}{v}$

Pg (11) **it's Further Applications:**

Atomizer

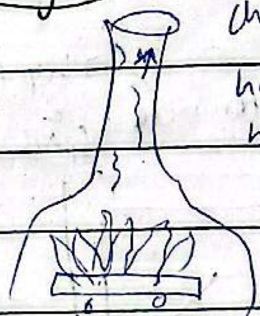
Apply force on pump → Air molecules speed increases → Inside pipe speed increases → pressure decreases → pressure has relatively decreased w.r.to container in which water is filled → Relative pressure inside liquid increases → Force will apply from high pressure to low pressure. → liquid level rises → & this liquid get sprayed in outward direction.

Drinking something from Straw

Straw pipe has smaller Area → when you suck liquid → due to smaller Area → V increases → P gets reduced → liquid pressure below will increase relatively → hence Force will be applied on liquid & liquid will rise from Straw.

Chimney → work best when there flow speedy air outside.

outside the house
V increase
P decrease
Air ke muqabale mai inside the room jahan par ap ne agg lagai huwi hai



Chimney upar se narrow down ho rahi huti hai ais ki waja se pressure difference create ho raha huta hai that help us in rising the smoke

Jab sard places par jate ho to outside hawayein chal rahi huti hai

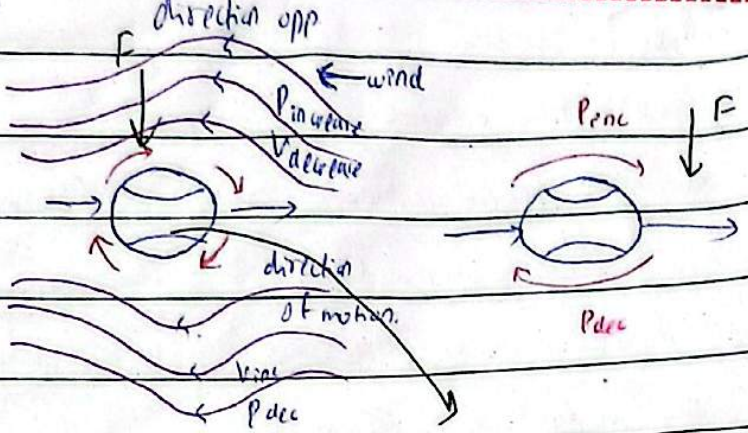
Pressure ~~relatively~~ increase huga
jis ki waja se smoke high p se low p ki taf jaye'gi.
(high p: pressure; low p: pressure)

Jis ki waja se smoke kamre se bahir chle jati hai

Pg (12)

Swinging Ball / Magnus Effect.

ball rotates as well as moves forward



uplift of Aeroplane (while lifting or landing of aeroplane.

while taking off Aeroplane open's Aeroplane planks in a way that below them air speeds decreases & at up V increases so due to this difference of speed. (upar koi opposition nahi mil rahi hoti)

At up → V inc., P dec

At down → V dec., P inc

Force will be applied from below to up that will help Aeroplane to take off.

(a) carburetor (b) Sail of Yatch (c) bunsen burner

(d) Filter pump

Venturi Relation:

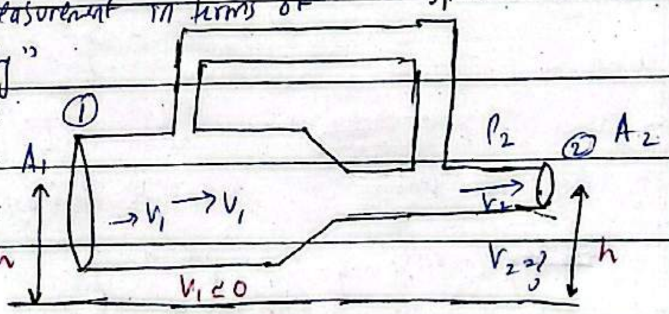
Venturi meter → help finding speed, flow rate.

~~"Speed of fluid measurement in terms of~~ Pressure & density"

For measuring speed we create pressure difference & find speed where pressure is low

~~P Dream~~

$P \propto 1/_V \propto A$



$\boxed{P_1 = P_2 + \frac{1}{2}\rho V_2^2}$ | $V_2 = \sqrt{\frac{2(P_1 - P_2)}{\rho}}$ | Venturi relation